Agent Generated Hypothesis and Research Plan

Novel Hypothesis:

Metformin, in combination with a MET inhibitor, synergistically impairs glutamine metabolism in Glioblastoma Multiforme (GBM) cells, leading to decreased cell proliferation and survival, particularly under nutrient-stressed conditions. This synergistic effect is mediated by AMPK activation, downstream inhibition of glutaminase, and a reduction in alpha-ketoglutarate production, ultimately disrupting the TCA cycle and promoting cell death.

Research Plan:

This preclinical research plan aims to test the hypothesis that Metformin, in combination with a MET inhibitor, synergistically impairs glutamine metabolism in GBM cells, leading to decreased cell proliferation and survival, particularly under nutrient-stressed conditions.

Phase 1: In Vitro Studies

- **Cell Lines:** Use a panel of GBM cell lines, including both patient-derived and established lines (e.g., U87-MG, A172, LN229). Include cell lines with varying levels of MET expression and glutamine dependence.
- **Treatment Groups:**
- Control (Vehicle)
- Metformin (clinically relevant concentrations, e.g., 1-10 mM)
- MET Inhibitor (e.g., Crizotinib, Tepotinib, Cabozantinib; concentrations determined based on IC50 values for the specific cell lines)
- Metformin + MET Inhibitor (Combination treatment at previously defined concentrations)
- **Cell Proliferation and Viability Assays:** Perform MTT or CellTiter-Glo assays to assess cell proliferation and viability at 24, 48, and 72 hours post-treatment. Compare the effects of single agents and the combination.
- **Glutamine Metabolism Assays:**
- **Glutaminase Activity Assay:** Measure glutaminase activity in cell lysates to assess the direct impact on glutamine breakdown.
- **Glutamate and Alpha-Ketoglutarate Quantification:** Use LC-MS or enzymatic assays to quantify intracellular levels of glutamate and alpha-ketoglutarate.
- **Tracing Experiments:** Use [13C5]-glutamine to trace glutamine metabolism through the TCA cycle. Measure the incorporation of 13C into downstream metabolites using mass spectrometry.
- **AMPK Activation Assay:** Measure AMPK phosphorylation (p-AMPK) via Western blot to confirm AMPK activation by Metformin. Also, examine downstream targets of AMPK, such as ACC (acetyl-CoA carboxylase).
- **Nutrient Stress Conditions:** Repeat the above experiments under glucose-deprived or glutamine-deprived conditions to mimic the tumor microenvironment. Assess whether the combination treatment is more effective under stress.
- **Apoptosis/Cell Death Assay:** Perform Annexin V/PI staining followed by flow cytometry to quantify apoptosis and cell death. Also, analyze caspase-3 cleavage by Western blot.

Phase 2: In Vivo Studies

- **Animal Model:** Use orthotopic xenograft models by injecting GBM cell lines (selected based on in vitro results) into the brains of immunocompromised mice (e.g., NSG mice).
- **Treatment Groups:**
- Control (Vehicle)
- Metformin (clinically relevant dose, e.g., 250 mg/kg/day via oral gavage)
- MET Inhibitor (dose determined based on previous in vivo studies)
- Metformin + MET Inhibitor (Combination treatment)
- **Tumor Growth Monitoring:** Monitor tumor growth using MRI or bioluminescence imaging (if using luciferase-expressing cells).
- **Survival Analysis:** Kaplan-Meier survival curves to assess the impact of treatments on overall survival.
- **Tissue Analysis:**
- **Immunohistochemistry:** Stain tumor sections for markers of proliferation (Ki-67), apoptosis (cleaved caspase-3), AMPK activation (p-AMPK), and glutamine metabolism enzymes (glutaminase).
- **Metabolomics:** Perform metabolomic analysis on tumor tissue to quantify the levels of glutamine, glutamate, alpha-ketoglutarate, and other relevant metabolites.

Phase 3: Mechanism of Action Studies

- **siRNA Knockdown:** Knockdown AMPK or glutaminase using siRNA in GBM cell lines. Assess the impact on cell proliferation, glutamine metabolism, and the synergistic effect of Metformin and the MET inhibitor.
- **CRISPR-Cas9 Knockout:** Generate AMPK or glutaminase knockout cell lines using CRISPR-Cas9. Confirm the loss of protein expression by Western blot. Assess the impact on cell proliferation, glutamine metabolism, and the synergistic effect of Metformin and the MET inhibitor.
- **Rescue Experiments:** Overexpress glutaminase in cells treated with Metformin and the MET inhibitor to determine if glutaminase overexpression can rescue the cells from the effects of the drugs.

Novelty Assessment:

Based on the PubMed searches performed, the use of Metformin in combination with other agents targeting GBM metabolism has been explored, specifically with dichloroacetate (DCA). Furthermore, there is evidence linking MET inhibition to metabolic reprogramming and altered glutamine metabolism in GBM. However, the *specific hypothesis* of a synergistic effect between Metformin and a MET inhibitor *directly targeting glutamine metabolism via AMPK activation* to impair GBM cell survival, particularly under nutrient stress, appears to be a novel combination. While individual components have been studied, the proposed mechanism and specific synergy have not been explicitly demonstrated based on the search results. Further literature review is always recommended prior to initiating the research.